

Austin Jiang

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EXPERIENCE

Noah's Ark Lab, Huawei Canada

Toronto, ON

Research Intern | Python, PyTorch

May 2026 – Present

- Designed an efficient 3D reconstruction model to reduce token redundancy in Visual Geometry Transformer.
- Introduced an **residual-update gating** method for adaptive token early-halt to freeze background computation.
- Achieved a **30% speedup** and **3% accuracy boost**, breaking the accuracy-compute Pareto frontier.
- Built a **3-stage desensitization training** method to jointly optimize gating layers and fine-tune the backbone.

Multicore Lab, University of Waterloo

Waterloo, ON

Undergraduate Research Assistant (Supervised by Dr. Trevor Brown) | C++, CUDA

Sep 2025 – Present

- Developed a unified adapter to support versioned and non-versioned data structures within the benchmark.
- Benchmarking Version Lib (PPoPP'24) extension with **10M-key** range queries under **100+ threads** on GPU.
- Built **B+ Tree** and **ART Tree** against a baseline **GPU B-Tree** using the GPU Range Query Dictionary library.

Wolfram Research

Remote

Research Intern (Part-time) | Mathematica

Sep 2025 – May 2026

- Developed multi-threaded engine managing **over 800K** data points across **8 CPU cores**, achieving **1.4x speedup**.
- Developed sparse frontier updates and dense scans, achieving a **127% speedup** via load-balancing strategies.
- Designed a **benchmark harness** measuring updates per second, active ratio, and scheduling overhead.

Research Intern (Part-time) | Mathematica

Sep 2024 – Jan 2025

- Developed algorithms to voxelize 3D meshes, detect overhanging regions, increasing infill stability by **55%**.
- Designed **benchmarking metrics** to measure density, connectivity, and printability of generated infills.
- Produced a first-author research paper, presented at the **Wolfram Technology Conference 2025**.

PROJECTS

Rewrite, TreeHacks 2026 | Python: Built a real-time 3D VR game in Unity that is fully interactive, deploying Hunyuan WorldPlay 1.5 on Modal across 4x H200 GPUs with tensor parallelism (accelerating from 1 fps to 12 fps).

ADaptiv, NexHacks 2026 | Python: Built a full video object replacement pipeline in ComfyUI, using Gemini for video understanding, Meta SAM3 for pixel-level masking, and Alibaba Wan, accelerated on RTX 4090.

LookAround AI, AdventureX (Multimodel Track Winner) | Python, React: Built a voice-controlled multi-agent tour guide using the TEN Framework, integrating Google Maps Street View API for route narration.

Personal Infrastructure & Services | Linux, FastAPI, SQL: Built & maintained a personal Linux server hosting a full-stack website behind Nginx, Cloudflare DNS, cloud storage, OpenVPN, email service, and FastAPI + SQL backends.

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science (Honours) | Major GPA: 4.0/4.0

Expected Graduation: Apr 2029

- Scholarships:** René Descartes National Scholarship (\$25,000), President's Scholarship of Distinction (\$2,500)
- Coursework:** Data Structures, Algorithms, Development Tools, Linear Algebra

SKILLS

Languages: C++, Python, C, Java, Bash, TypeScript, Rust

Tools/Libraries: Pytorch, Linux, Git, CUDA, CMake, Numpy, SQL, Docker, Nginx, Flask, FastAPI

Concepts: Transformers, Large Language Models, Concurrency, Compilers, Distributed Systems, Database

ACHIEVEMENTS

Canadian Computing Olympiad 2026: Authored Day 1 Problem 3 (Contestant average score: 0.7/25).

Discover Citadel Program: Invited as 1 of ~100 students (< 1% acceptance rate).

Jane Street First-Year Trading and Technology Program: Invited as 1 of ~100 students (< 1% acceptance rate).

Generation Google Scholarship: Google's flagship global undergrad scholarship for impact in technology (1 of 55).

Canadian Computing Olympiad 2024 & 2025: Silver medalist x2 (5th and 6th out of 10,000+ participants).

USACO 2024 (Platinum): Achieved the highest division, ranked top 100 out of 15,564 participants.